# Questions

Q1.

A particle $P$ has mass $km$ and a particle $Q$ has mass $m$. The particles are moving towards each other in opposite directions along the same straight line when they collide directly.

Immediately before the collision, $P$ has speed $3u$ and $Q$ has speed $u$.

As a result of the collision, the direction of motion of each particle is reversed and the speed of each particle is halved.

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(a) Find the value of $k$.

**(4)**

(b) Find, in terms of $m$ and $u$, the magnitude of the impulse exerted on $Q$ in the collision.

**(3)**

**(Total for question = 7 marks)**

(**Q02 WME01/01, Jan 2022**)

Q2.

A particle $P$ has mass $3m$ and a particle $Q$ has mass $5m$. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

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Immediately before the collision the speed of $P$ is $ku$, where $k$ is a constant, and the speed of $Q$ is $2u$.

Immediately after the collision the speed of $P$ is $u$ and the speed of $Q$ is $3u$.

The direction of motion of $Q$ is reversed by the collision.

(a) Find, in terms of $m$ and $u$, the magnitude of the impulse exerted on $Q$ by $P$ in the collision.

**(2)**

(b) Find the two possible values of $k$.

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**(5)**

**(Total for question = 7 marks)**

**(Q01 WME01/01, June 2021)**

Q3.

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Two particles, $P$ and $Q$, have masses $2m$ and $m$ respectively. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal plane. The particles collide directly.

Immediately before the collision, the speed of $P$ is $3u$ and the speed of $Q$ is $2u$.

The magnitude of the impulse exerted on $Q$ by $P$ in the collision is $5mu$.

Find

(a) the speed of $P$ immediately after the collision,

**(3)**

(b) the speed of $Q$ immediately after the collision.

**(3)**

**(Total for question = 6 marks)**

**(Q02 WME01/01, Jan 2021)**

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Q4.

Two particles, $P$ and $Q$, are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision the speed of $Q$ is $2u$. The mass of $Q$ is $3m$ and the magnitude of the impulse exerted by $P$ on $Q$ in the collision is $4mu$.

Find

(a) the speed of $Q$ immediately after the collision,

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**(3)**

(b) the direction of motion of $Q$ immediately after the collision.

**(1)**

**(Total for question = 4 marks)**

**(Q01 WME01/01, June 2022)**

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Q5.

A railway truck $S$ of mass 20 tonnes is moving along a straight horizontal track when it collides with another railway truck $T$ of mass 30 tonnes which is at rest. Immediately before the collision the speed of $S$ is 4 m $s^{-1}$

As a result of the collision, the two railway trucks join together.

Find

(a) the common speed of the railway trucks immediately after the collision,

**(2)**

(b) the magnitude of the impulse exerted on $S$ in the collision, stating the units of your answer.

**(3)**

**(Total for question = 5 marks)**

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**(Q01 WME01/01, Oct 2022)**

Q6.

A particle $P$ of mass $2m$ is moving on a rough horizontal plane when it collides directly with a particle $Q$ of mass $4m$ which is at rest on the plane. The speed of $P$ immediately before the collision is $3u$. The speed of $Q$ immediately after the collision is $2u$.

(a) Find, in terms of $u$, the speed of $P$ immediately after the collision.

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**(3)**

(b) State clearly the direction of motion of $P$ immediately after the collision.

**(1)**

Following the collision, $Q$ comes to rest after travelling a distance $\frac{6u^2}{g}$ along the plane.

The coefficient of friction between $Q$ and the plane is $\mu$.

(c) Find the value of $\mu$.

**(6)**

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**(Total for question = 10 marks)**

(**Q02 WME01/01, Oct 2021**)

Q7.

A hammer is used to hit a tent peg into soft ground.

The hammer has mass 1.8 kg and the tent peg has mass 0.2 kg.

The hammer and tent peg are both modelled as particles and the impact is modelled as a direct collision.

Immediately before the impact, the tent peg is stationary and the hammer is moving vertically downwards with
speed 10 m $s^{-1}$

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed $v$ m $s^{-1}$

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(a) Find the value of $v$

**(2)**

(b) Find the magnitude of the impulse exerted on the tent peg by the hammer, stating the units of your answer.

**(3)**

The ground exerts a constant vertical resistive force of magnitude $R$ newtons, bringing the hammer and tent peg to rest after they travel a distance of 12 cm.

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(c) Find the value of $R$.

**(5)**

**(Total for question = 10 marks)**

(**Q03 WME01/01, Oct 2023**)

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Q8.

A particle $A$ has mass 4 kg and a particle $B$ has mass 2 kg.

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of $A$ is $2u$ m $s^{-1}$ and the speed of $B$ is $3u$ m $s^{-1}$

Immediately after the collision, the speed of $B$ is $2u$ m $s^{-1}$

The direction of motion of $B$ is reversed by the collision.

(a) Find, in terms of $u$, the speed of $A$ immediately after the collision.

**(3)**

(b) State the direction of motion of $A$ immediately after the collision.

**(1)**

(c) Find, in terms of $u$, the magnitude of the impulse received by $B$ in the collision. State the units of your answer.

**(3)**

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**(Total for question = 7 marks)**

(**Q01 WME01/01, June 2023**)

Q9.

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A block $A$ of mass 9 kg is released from rest from a point $P$ which is a height $h$ metres above horizontal soft ground. The block falls and strikes another block $B$ of mass 1.5 kg which is on the ground vertically below $P$. The speed of $A$ immediately before it strikes $B$ is 7 m $s^{-1}$. The blocks are modelled as particles.

(a) Find the value of $h$.

**(2)**

Immediately after the impact the blocks move downwards together with the same speed and both come to rest after sinking a vertical distance of 12 cm into the ground. Assuming that the resistance offered by the ground has constant magnitude $R$ newtons,

(b) find the value of $R$.

**(8)**

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**(Total for question = 10 marks)**

(**Q03 WME11/01, Specimen papers**)

Q10.

Two particles, $A$ and $B$, are moving in a straight line in opposite directions towards each other on a smooth horizontal surface when they collide directly.

Particle $A$ has mass $3m$ kg and particle $B$ has mass $m$ kg.

Immediately before the collision, both particles have a speed of 1.5 m $s^{-1}$

Immediately after the collision, the direction of motion of $A$ is unchanged and the difference between the speed of $A$ and speed of $B$ is 1 $m\ s^{-1}$

(a) Find (i) the speed of $A$ immediately after the collision,

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(ii) the speed of $B$ immediately after the collision.

**(5)**

(b) Find, in terms of $m$, the magnitude of the impulse exerted on $B$ in the collision.

**(3)**

**(Total for question = 8 marks)**

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(**Q02 WME01/01, Jan 2023**)

Q11.

Two particles $P$ and $Q$ are moving in opposite directions along the same horizontal straight line. Particle $P$ has mass $m$ and particle $Q$ has mass $km$. The particles collide directly. Immediately before the collision, the speed of $P$ is $u$ and the speed of $Q$ is $2u$. As a result of the collision, the direction of motion of each particle is reversed and the speed of each particle is halved.

(a) Find the value of $k$.

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**(4)**

(b) Find, in terms of $m$ and $u$ only, the magnitude of the impulse exerted on $Q$ by $P$ in the collision.

**(2)**

**(Total for question = 6 marks)**

(**Q02 WME11/01, Specimen papers**)

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